

ELECTIVE COURSE-3.4

URA743: AUTOMOTIVE MECHATRONIC SYSTEMS				
	L	T	P	Cr
	3	0	0	3.0
Course Objective: This course is designed to provide you with a comprehensive understanding of Human-Machine Interfaces (HMIs). You will learn about the architecture and components of HMIs, develop skills in 3D game development using Unity 3D, create HMI applications using the QT framework, and gain expertise in Android and web app development. Additionally, you will explore the challenges and best practices for HMI testing and automation.				
Syllabus Automotive Electronics: Intervention of Electronics in Vehicle Operation, Vehicle System Architecture, Electronic Control Units, Data Processing, Control Mechanism, Requirements & Classifications of BUS system, CAN Bus, LIN Bus, Network Topologies, Diagnostic Interface. Introduction to body computers, body computer module, microprocessors, highside drivers, low-side drivers. Automotive Sensors and Measurements: Automotive Sensor requirements, Smart Sensors, Automotive sensors like- angle, position, speed, acceleration, torque, fuel consumption measurement, Lambda oxygen sensor, iBolt force sensor, Rain, light sensor. Electronic Control of Combustion: Electronic combustion control; control of exhaust gas recirculation; electronic control of valve timing; distributorless ignition systems (DIS); fuel metering and control; electronic fuel injection: single point, multi point and direct injection. Advanced Automotive Lighting: Computer controlled headlight systems; automatic on/off with time delay; automatic headlight dimming; headlight levelling; adaptive headlights; adaptive Brake Lights; instrument panel dimming; lamp outage indicators; LED lamps; cornering lights. Driver Assistance System: Digital instrument cluster; travel information system; headupdisplay; night vision system; global positioning navigation system; lane change warning system; warning programs; traffic management system; hand's free communication and operation. Automatic Transmission: Drive by wire system; electronic shift transmission; direct shift gearbox; paddle shift control; constantly variable transmission; cruise control. Active and Passive Safety Systems: Introduction; primary restraint system; secondary restraint system; passive seat belt systems; air bag systems; seat belt pre-tensioners; inflatable knee blockers; occupant classification systems; anti-whiplash headrest restraint system; antilock braking system. On-Board Diagnosis (OBD): Vehicle management system; requirement of OBD-II; fault diagnosis; fault codes; vehicle fault scanners; vehicle oscilloscope.				
Laboratory Work Programming exercises for development of HMI in Unity and development of UI/UX using protopie.				
Course Learning Objectives (CLO) The students will be able to: 1. correlate the function of electronic systems in modern automobiles.				

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2. evaluate the use of electronics to automate mechanical controls for improved performance, safety, comfort and related issues.
3. specify the intervention of electronic devices in saving fuel by controlling the combustion process and pollutants.

Text Books

1. Automotive Mechatronics: Automotive Networking, Driving Stability, Systems, Electronics, Konrad Reif, Bosch Professional Series, Springer,
2. Hollembeak, B., Today's Technician: Advanced Automotive Electronic Systems, 6th Edition, Delmar Cengage Learning, USA (2014).
3. Hiller, V. A. W., Fundamentals of Automotive Electronics, 6th Edition, Oxford University Press, UK (2014).
4. Hiller, V. A. W., Fundamentals of Motor Vehicle Technology, 6th Edition, Oxford University Press, UK (2012).

Reference Books & Websites

1. Heisler, H., Advanced Vehicle Technology, 2nd Edition, Butterworth-Heinemann (2002)
2. Pike, J. A., Automotive Safety, 1st Edition, SAE International (2009).
3. Duffy, J.E., Modern Automotive Technology, 9th Edition, Goodheart-Willcox Publisher (2016).

Evaluation Scheme:

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Sr. No.	Evaluation Elements	Weightage (%)
1	MST	25
2	EST	40
	Evaluation Elements	Weightage (%)

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